Mapping Best Empirical Models

A Meta-Analysis of Winning Solutions in Kaggle Tabular Competitions

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Motivation

The folklore

Winning a tabular Kaggle competition is supposed to be about elaborate feature engineering.

The clever, hand-crafted features that separate the best from the rest.

But the winners disagree

Among the winners:

4	6	62%
median own features (best-documented winners)	winners engineered *no* features	simply ensembled existing models

Winning isn't the feature-engineering craft the folklore promises.

What we “know” is anecdotal

- Folk wisdom lives in blog posts, forum threads, and winner interviews.
- Academic benchmarks compare algorithms on fixed datasets, never winners against strong non-winners.
- Nobody has systematically asked what *distinguishes* a winning solution from a near-winning one.

This study codes 45 winning solutions + 11 near-winning controls.

Research questions

RQ1 What strategy paradigms define winning? Do folklore “constraint $\rightarrow$ strategy” rules hold?

RQ2 Which feature-engineering techniques do winners use, and how much?

RQ3 Does feature engineering distinguish winners from strong non-winners?

RQ4 How is the winning population structured?

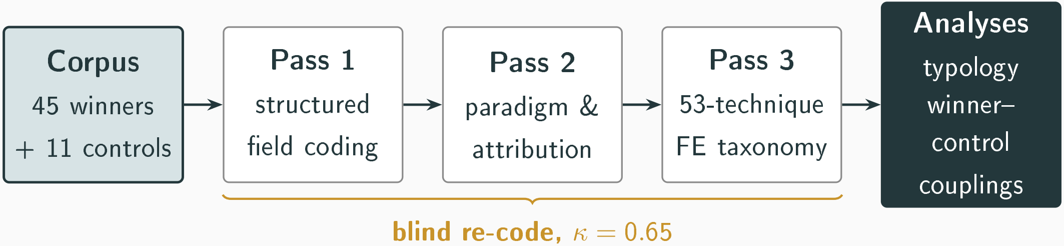
Data & Method

The corpus

- 45 winning solutions (top-three finishers)
- 11 near-winning controls (ranks 4–25, same competitions)
- February 2022 – April 2026
- Mostly Kaggle Playground Series (synthetic data); one real-data Featured competition
- Coded on a 40-column schema plus a 53-technique feature-engineering taxonomy
- Caveat: only solutions with public write-ups can be coded (documentation self-selection)

A content-analysis meta-study

No model is trained: the unit of analysis is one completed solution, coded in three passes.



Each winner is paired against a near-winning control from the *same* competition, isolating what distinguishes winning from merely strong play.

Checking our own coding

- Primary coding was AI-assisted, researcher-directed.
- A blind re-code of a stratified 12-entry subsample gives $\kappa = 0.65$: substantial agreement at the technique-family level.
- So FE results are reported at the *family* level, where they are reliable.

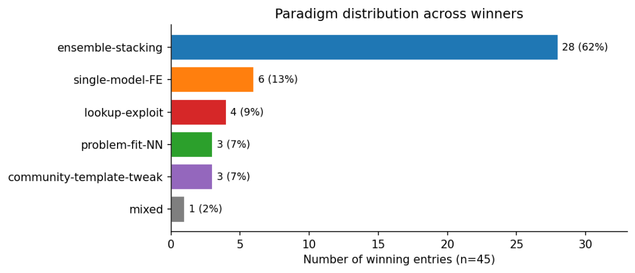
Limitations: small *n* (11 paired competitions); synthetic-data corpus; intra-rater (not inter-rater) reliability; documentation self-selection.

Findings

What winners actually use

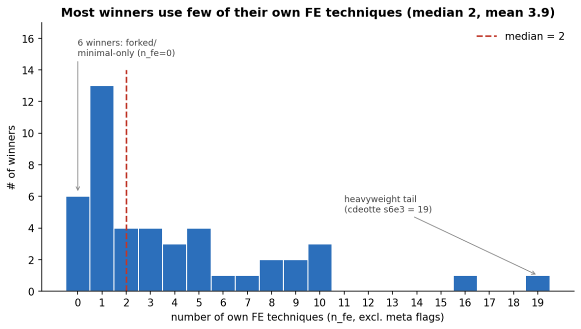
A few models (GBMs) and ensemble stacking dominate; FE is a long, varied tail.

Winning reduces to a few paradigms



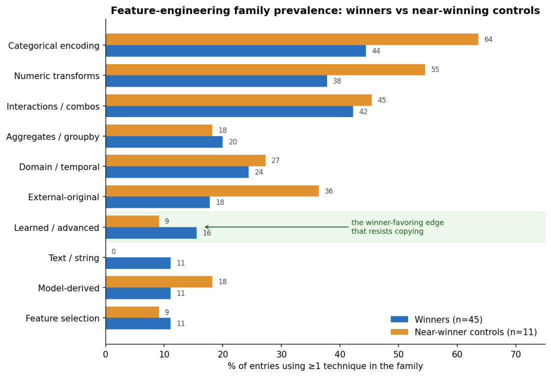
- Ensemble stacking = 62% of wins (28/45)
- Gradient-boosted trees are the dominant model in 35 of 45
- A short tail: single-model-FE, lookup-exploit, problem-fit-NN

Winning FE is a small shared core



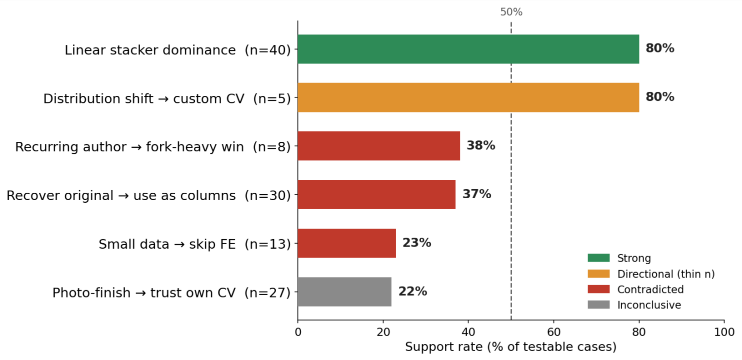
Even the best-documented winners engineer a median of 4 of their own features (a corpus-wide floor of 2), and 6 used none. Target encoding is the one near-universal technique; the rest is a long tail.

FE does not separate winners from near-winners



- Aggregate: controls do *more* FE, but that's a confound (their competitions are FE-heavy)
- Paired, within-competition: winners do more in 7 of 11 (n.s.)
- Learned features are the one family that consistently leans winner

Folklore “rules” mostly don’t hold

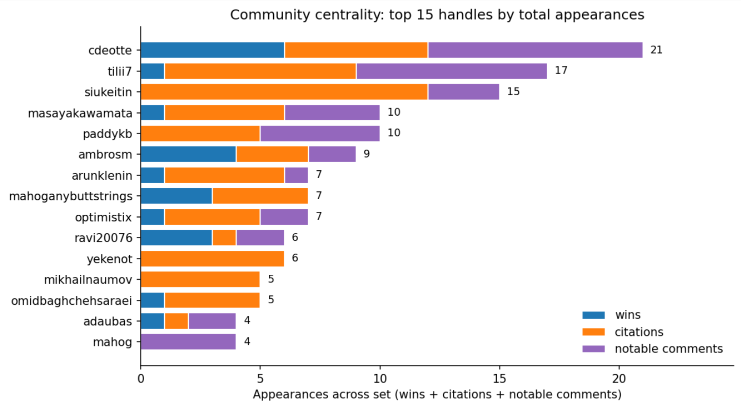


The lone green survivor, linear-stacker dominance, just restates standard ensembling.
The 80% “custom CV” rule rests on only 5 cases.

The documented top is a recurring guild

45%

author overlap between winners and near-winners



A few central figures supply widely-used techniques without always winning (originators vs. canonizers); recognition accrues to the already-prominent.

Conclusions

Three takeaways

1. Winning $\neq$ FE folklore: a small shared core, not elaborate craft.
2. Learned features are the one feature-level edge that resists copying from a public notebook.
3. Winning is, in part, a community phenomenon.

What this does, and does not, establish

- Claims are scoped to the documented competitive elite: not typical competitors, not real-world tabular ML.
- The corpus is mostly synthetic Playground data; some strategies are generator artifacts.
- The winner–control test is underpowered ($n = 11$): the result is "no clean separation," not "no difference."

Contributions & future work

Contributions

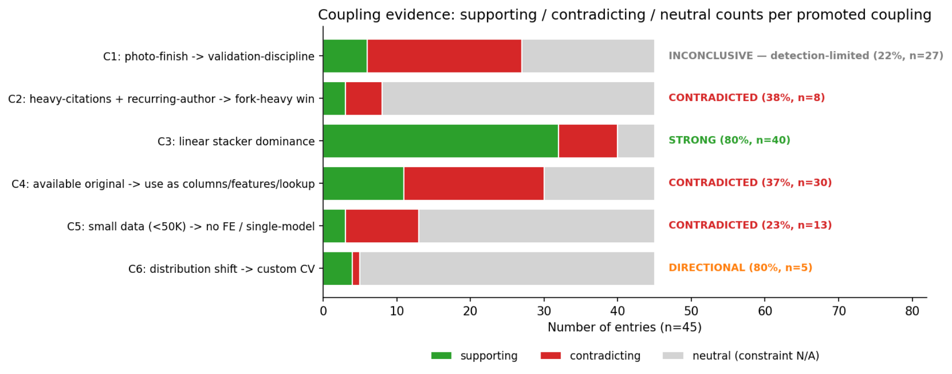
- The first structured, reliability-checked characterization of winners *versus near-winners*.
- A reusable 53-technique FE taxonomy and multi-pass coding workflow.
- Replaces folklore with documented frequencies.

Future work

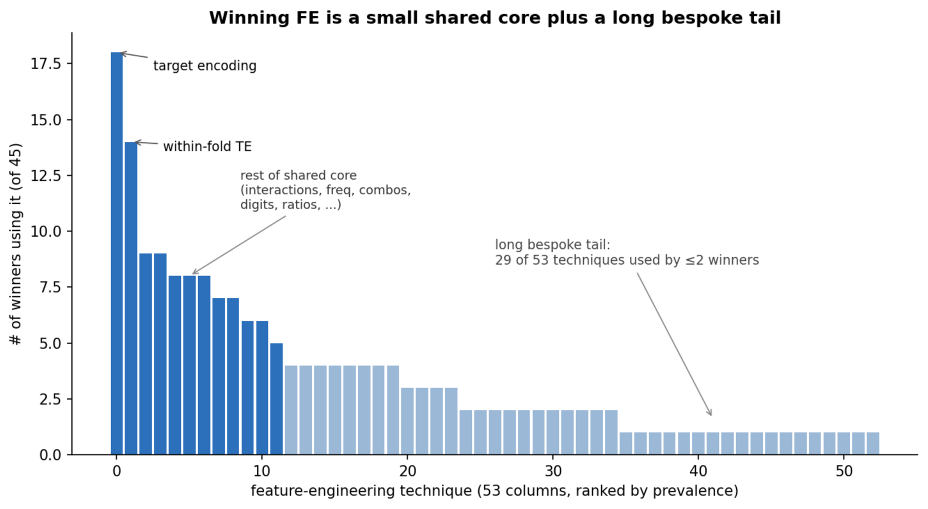
- Track the emerging Season-6 shift: very large (100+ model) ensembles and LLM-assisted workflows the four-paradigm typology will need to absorb.
- Real-data Featured competitions; other platforms and modalities.

Winning turns less on the *volume* of feature engineering than on paradigm fit and who competes.

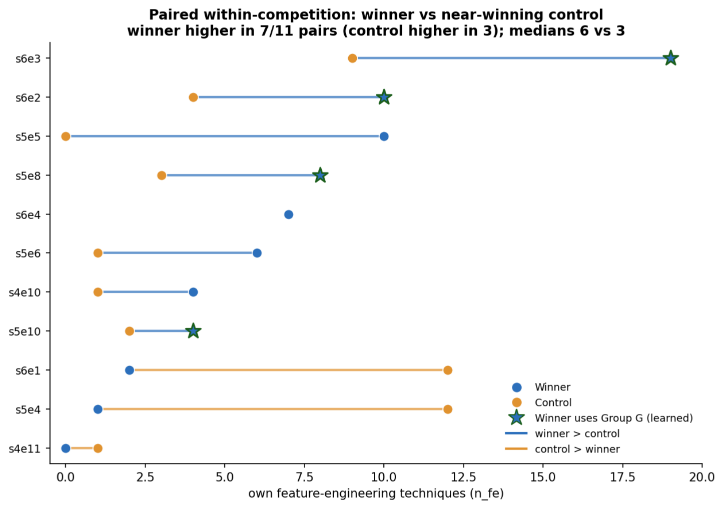
Appendix: coupling evidence (detailed)



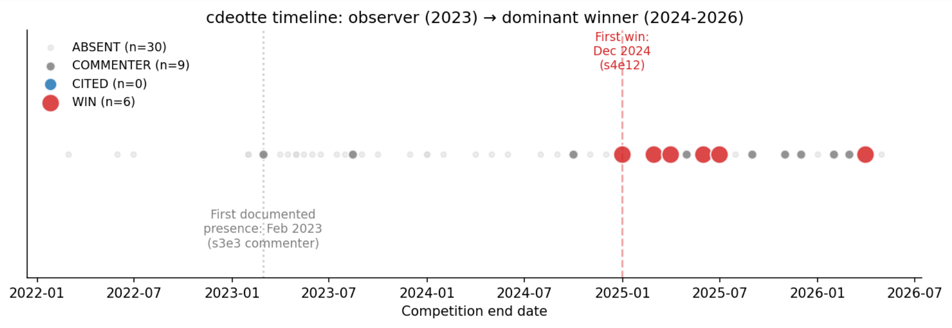
Appendix: feature-engineering technique long tail



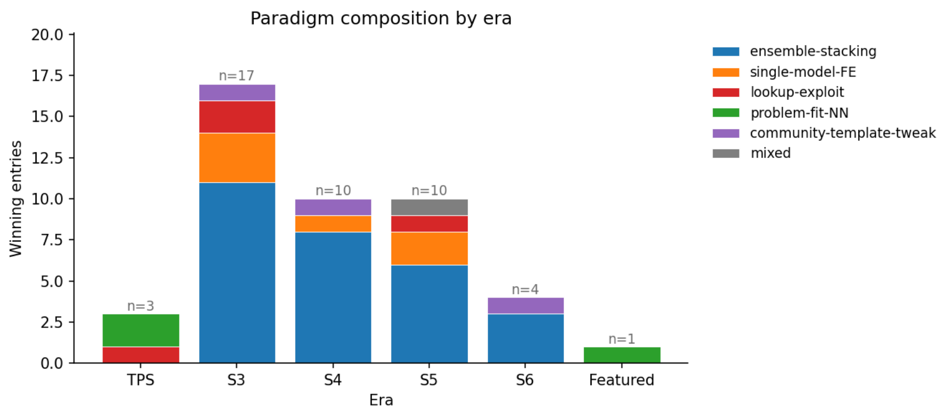
Appendix: paired winner vs. control (n_{fe})



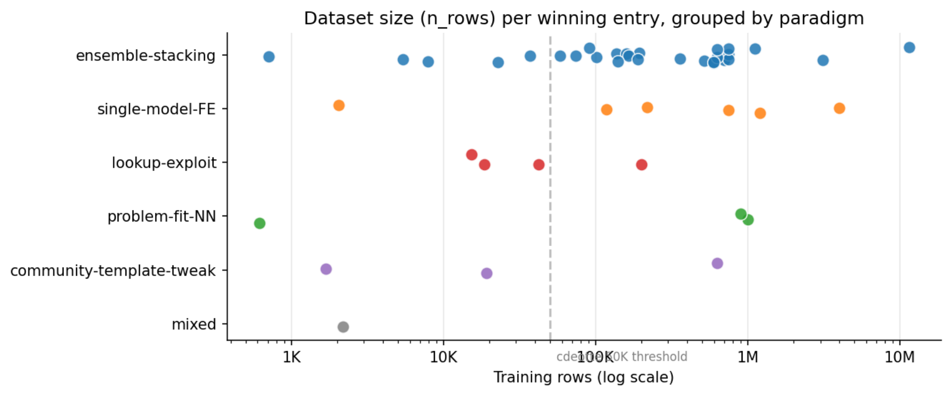
Appendix: cdeotte, observer to dominant winner



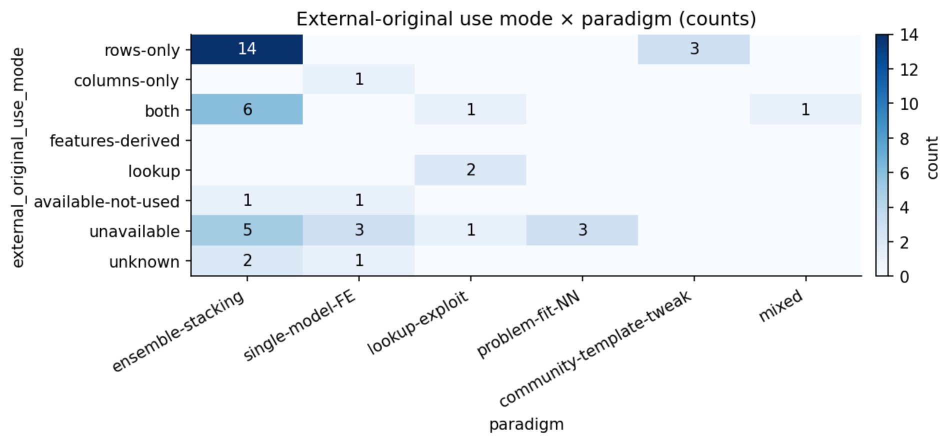
Appendix: paradigm composition by era



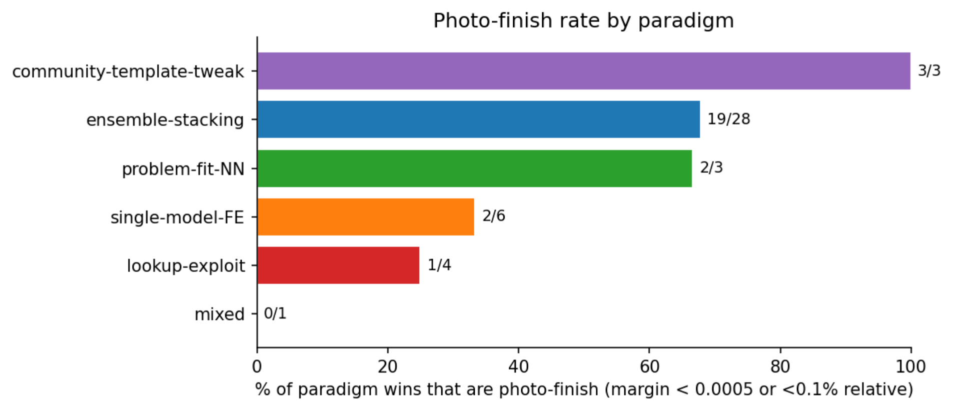
Appendix: dataset size by paradigm



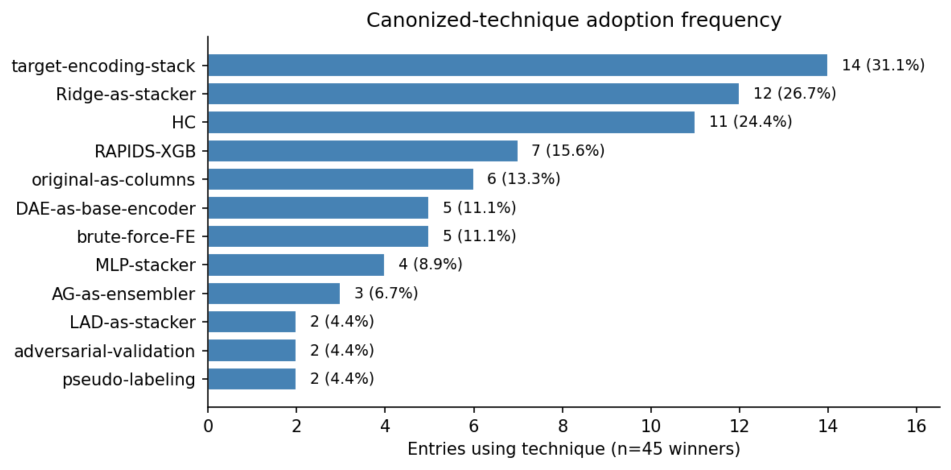
Appendix: external-original use by paradigm



Appendix: photo-finish rate by paradigm



Appendix: most-canonized techniques



Appendix: citations by origination score